

CLASS 09 / OCT 20 / U05

PLOTTER WORKFLOW AND SITE REVIEW

P3c + P3d CHECKPOINT — FIELD TO LINES, SITE TO DECISION

NOTHING DUE

MIDTERM NEXT CLASS — WEEK 10 — OCT 27

A PLOTTER DRAWS LINES, NOT PIXELS

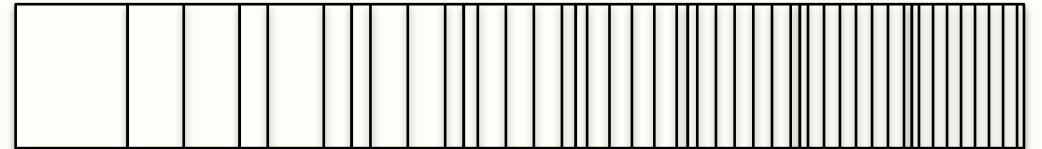
A screen can show any grey. A pen is either down or up.

ON SCREEN — A VALUE PER PIXEL



Every sample can be any grey. Nothing needs to be drawn.

ON PAPER — A STROKE OR NOTHING



Every stroke is the same darkness. Grey comes from how many lines, how long and how close.

SO THE FIELD MUST BECOME GEOMETRY: CURVES WITH A START AND AN END. AN IMAGE WILL NOT PLOT.

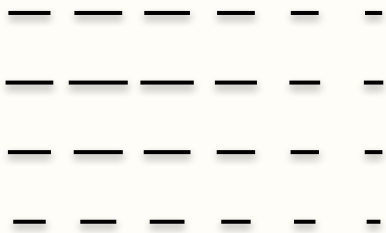
The grayscale field stays on the sheet as the legend. The line drawing is what the pen draws.

TRANSLATE THE FIELD INTO LINES

Same field, same remap. Only what the output number drives changes. Pick one.

LINE LENGTH

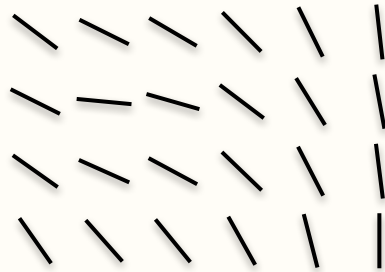
length = out



Long near, short far.

ROTATION

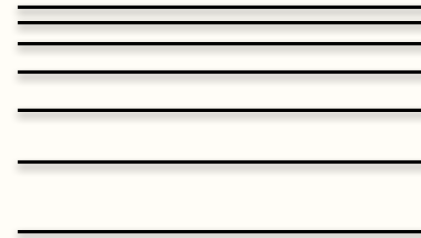
angle = out



Lines turn with distance.

SPACING

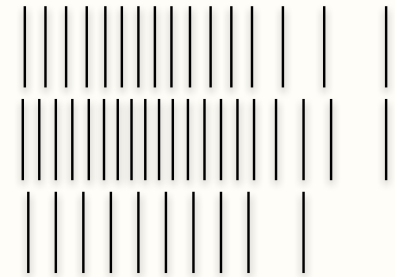
gap = out



Rows open up with distance.

HATCH DENSITY

lines per cell = out



More strokes, darker cell.

```
t = remap(distance(p, a))      # the field, unchanged
line(p, angle, length)         # the last line is the translation
```

KEEP IT A CURVE. EXPORT AT SIZE.

The SVG is the drawing the plotter receives. Check it before the pen touches paper.

CURVES

Keep strokes as curves to the end. Do not convert them to a mesh or a filled shape.

SIZE

Set the drawing to the plotted size — 17 × 11 inches, or the confirmed plotter size — and export at 100%.

UNITS

Know the unit of the file and of the export. Measure one known line in the SVG.

PEN

One pen weight, or a declared set of pens, one layer or colour per pen.

MARGINS

Keep every stroke inside the paper, with a margin the pen can reach.

OPEN THE EXPORTED SVG ON ITS OWN AND MEASURE IT. WHAT YOU SEE THERE IS WHAT WILL PLOT.

TEST PLOT SMALL FIRST

A small crop of the real drawing, on the real paper, with the real pen.

01 CROP

Take a small, dense corner of the drawing. Same line weight, same spacing.

02 PEN

Check the tip and the ink. A dry or worn pen skips.

03 PAPER

Fixed flat, the right size, the right way round.

04 SPEED

Slower for dense areas. Watch the first strokes.

THEN COMPARE THE TEST WITH THE SCREEN

Does the darkest area still read as separate lines? Does the lightest area still read at all? Change the mapping, not the pen.

P3c CHECKPOINT TODAY: A SMALL TEST PLOT, IN YOUR HAND.

COMMON PROBLEMS

Each one is visible in the SVG before it is visible on paper.

DUPLICATE LINES

Two strokes on top of each other. The pen draws both: darker ink, longer plot, weaker paper.

Remove overlaps before export.

TOO-DENSE HATCHING

Lines closer than the pen is wide merge into a blot and can wet through the paper.

Keep spacing above the pen width.

FILLS

A filled shape has no stroke to follow. It will not plot, or it plots only its outline.

Turn fills into hatching.

TOO MANY TINY STROKES

Thousands of very short segments make a slow plot with little to see.

Simplify, or lower the count.

IF IT LOOKS WRONG AT 100% ON SCREEN, IT WILL LOOK WORSE ON PAPER.

MIXTLI WORKING REVIEW — THE SITE

Mixtli is the Blender add-on provided for site analysis. Before any analysis, record the site.

SOURCE

Where the site model or data came from. Name it on the sheet.

UNITS

What one unit is. Check it against one known dimension.

SCALE

The scale of the site plan or base view, with a bar.

ORIENTATION

North, and how the model is placed relative to it.

AND ONE SENTENCE: WHY THIS SITE

A site you can document, and a reason you chose it. The reason becomes your analysis question.

P3d CHECKPOINT TODAY: YOUR SITE, ONE ANALYSIS WITH A LEGEND, AND A QUESTION.

ONE ANALYSIS, ONE DECISION

The add-on does the computation. The question, the comparison and the reading are yours.

01 ANALYZE

One analysis. Record its type, inputs and settings. Legend and units on the image.

02 COMPARE

Change one thing — a setting, a time or a condition — and show both.

03 INTERPRET

Say what the output measures. Read it as a field: a value at every point, remapped to colour.

04 DECIDE

Name one spatial decision the reading supports, and draw it on the plan.

SAME LOGIC AS P3c: A MEASURED VALUE, A STATED RANGE, A LEGEND, THEN A CONSEQUENCE.

A colourful image with no legend and no decision is not an analysis.

MIDTERM CHECKLIST — PROJECTS 1–3

Next class. Due then: GSM · MID · P2b · P3c · P3d. Tick every line before you print.

01

Print and present Project 1: the four-page Graphic Standards Manual.

02

Present Project 2: the CSG process sheet, the 3D printed CSG object and its process sheet.

03

Present Project 3: one module sheet, two array sheets, the plotted attractor field drawing and one Mixtli site-analysis sheet.

04

Bring editable files and dependencies. Change a parameter live, predict the result and explain the nodes or the Mixtli analysis settings.

05

The review assesses revision and coherence across Projects 1-3. Earlier assignment grades remain separate. Project 4 follows the review.

NOW — CHECKPOINTS

Rest of the class: one table at a time, then work on what the review found.



P3c: grayscale field with a legend, beside the line translation



P3c: an SVG at plotted size, and a small test plot



P3d: site recorded — source, units, scale, orientation



P3d: one Mixtli analysis with a legend and units, and a stated question



Midterm: every item on the checklist located, printed or booked for printing

FIX THE MAPPING TODAY. NEXT WEEK THERE IS NO TIME TO REPLOT.

BEFORE NEXT CLASS

MIDTERM

Week 10 — Oct 27 is the midterm review of Projects 1–3. Due then: GSM, MID, P2b, P3c, P3d.

PLOT

Plot the final P3c drawing at full size. Keep the SVG, a photograph or scan of the plot and the editable graph.

SHEET

Finish the P3d sheet: annotated site, mapped analysis with legend and units, one comparison, one decision.

BRING

Printed sheets, the plotted drawing, the 3D print, and editable files with dependencies. Be ready to change a parameter live.